Compact lunar image matching

Frozen south-pole render study | Independent project; no NASA/JPL affiliation

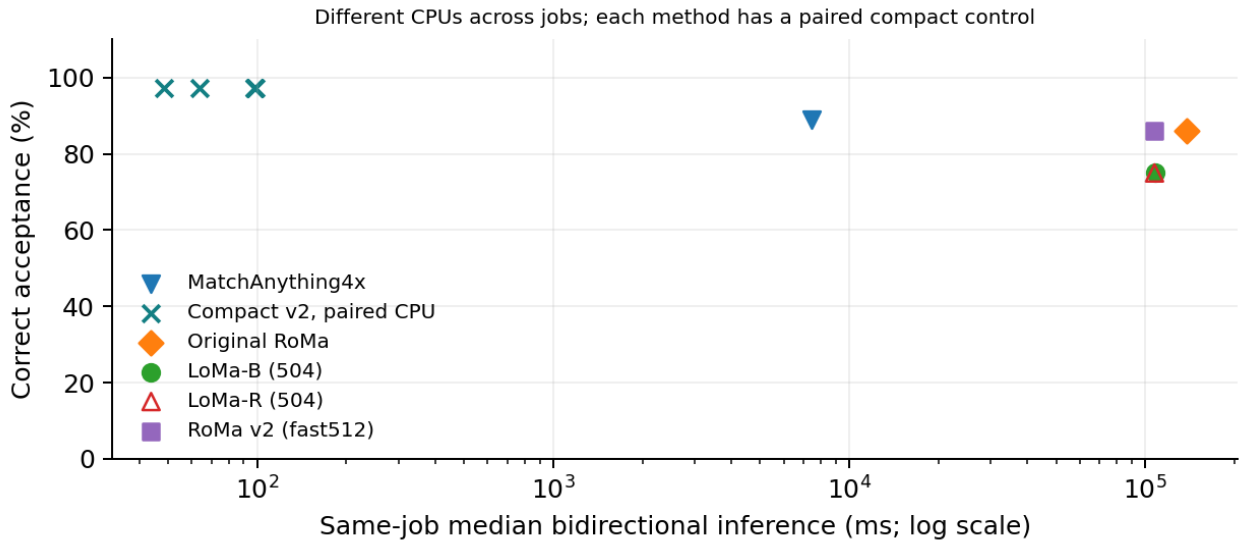
Question. Can a small illumination-aware matcher accept more correct alignments while rejecting incorrect matches, with a useful CPU tradeoff against larger models?

Design. 28 spatial components: 8 training, 4 model selection, 4 confidence calibration and 12 final test across three polar sectors. Configuration: expanded. Primary seed 20261030; all three predefined seeds retained. Reference geometry comes from simulated depth and cameras. Render correctness uses 3 input pixels, equivalent to 12 original 512px source pixels.

Ablations. Development raw correct positives: expanded: 93/96; reliability: 92/96; hard_negatives: 89/96; multiscale: 87/96. The selected architecture has 28,272 parameters. Added complexity did not improve this development criterion. Training used 18 polar geometries/54 lighting images.

Full frozen render test	Correct accepted	Wrong accepted	Negative accepted
Original CNN / frozen gate	100/384	0	0/288
Original CNN / new confidence	368/384	0	0/288
Compact v2 / primary	376/384	0	0/288
Compact v2 / seed 20261031	351/384	0	0/288
Compact v2 / seed 20261032	380/384	1	1/288

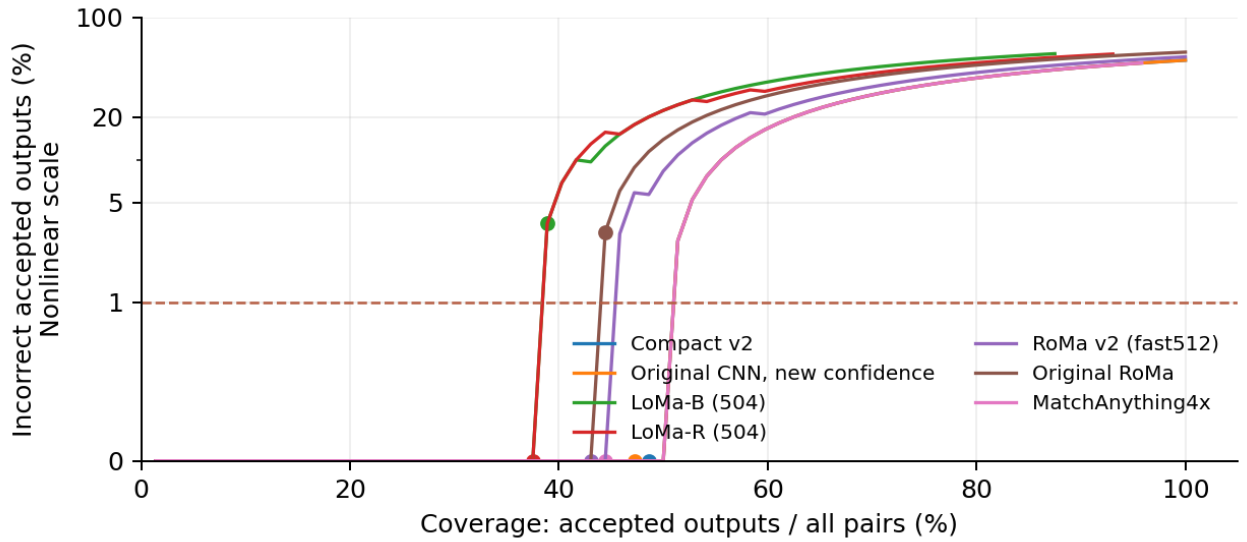
Measured gain. +71.88 percentage points versus the original frozen gate; exploratory grouped 95% interval +68.63 to +75.00. Predeclared 10-point/error-budget target met: True. Shared terrain/rendering dependence limits interpretation.



Runtime comparisons use paired same-job CPU controls only. Common modern comparisons contain 36 positives and 36 negatives; these are not published-paper benchmark percentages.

Comparison, transfer and limits

Identical common test	Raw correct	Correct accepted	Wrong accepted
Compact v2	36/36	35/36	0
Original CNN, new confidence	36/36	34/36	0
SIFT	0/36	0/36	0
LoMa-B (504)	28/36	27/36	1
LoMa-R (504)	30/36	27/36	0
RoMa v2 (fast512)	34/36	31/36	0
Original RoMa	31/36	31/36	1
MatchAnything4x	36/36	32/36	0



3D diagnostic. Compact estimates 142/288 poses; 2 meet the declared secondary 2-degree/1%-range criterion. Raw correspondences precede planar fitting; these simulated diagnostics do not establish navigation accuracy.

Sensor transfer. Compact: 23/40 nominally correct acceptances, 1 outside criterion. New NASA POLAR lighting configurations reuse inspected rocks/rig; reference drift uncertainty is unknown. Orbital navigation accuracy remains unverified.

Units. Inference 128px. The 3px render tolerance equals 12 original 512px pixels; POLAR nominal 3px equals 28.5 sensor pixels. These are coarse-registration diagnostics.

Limits. Calibration has only 12 positives and 12 negatives across 4 components; an observed 1% error budget is not a deployment guarantee. Negative pairs reuse both endpoints. Trajectory IDs and Sun angles are absent; selected spatial footprints do not establish trajectory independence. CPU allocations vary; speed comparisons use paired controls. Risk curves are descriptive; operating points were fixed before test. No global state-of-the-art or flight-readiness claim.

Sources: MoonAnything DOI 10.5281/zenodo.18415119; LoMa (ECCV 2026); RoMa v2, arXiv 2511.15706; NASA Ames POLAR. Repository contains case-level results, audits, licenses and reproduction commands.